

Final Project: The Meridian's Last Transmission

Your Name Here

A microbe called *Cinderspore* is spreading between star systems and dimming stars as it goes. One research vessel, the *Meridian*, was sent out with a small crew to study it and find a way to stop it before it reaches Earth.

The crew has one last transmission for you. Mission Control needs your analysis before deciding what to do next.

Final Mission Log — from Frank

Analyst. Hi. Usually I have a whole introduction, but we are a little short on time and the comms array is doing something I would describe as expressive.

We found a promising containment treatment. That is the good news. Before Command commits to the next stage, they want a final report on the treatment, the equipment, and the crew. I pulled the records.

Some of these questions sound similar. They are not all asking the same thing. Please read carefully. Also, if the data disagrees with me, tell Command. I would rather be wrong than have the ship be wrong.

Your Report

Answer all eight questions using methods covered in this course. You are responsible for choosing and justifying the analysis for each question.

Keep the report focused. Most questions should need one analysis and a short explanation.

The records are from separate trials or different crew members unless a question explicitly states that measurements are linked. You may treat separate observations as independent.

Data Import

Call in any packages you need and import the supplied datasets.

Question 1: Is Containment Strong Enough?

Frank: "The treatment is supposed to kill more than 80% of the *Cinderspore* in a standard culture. Science tested it on 40 separate cultures. Command wants evidence that the population mean exceeds 80%, not just one particularly impressive petri dish."

Use the containment dataset.

Variable	Description
killed_pct	Percentage of Cinderspore killed in a culture

Describe the treatment’s performance using appropriate numerical summaries and one graph. Estimate the population mean percentage killed with a confidence interval, and evaluate Command’s claim. Explain any assumptions needed for your analysis.

Your analysis and interpretation here.

Question 2: Can We Trust the Emergency Beacons?

Frank: “We tested 100 independently selected emergency beacons. A beacon either transmitted successfully or it did not. Fleet policy requires a success rate of at least 95%. I am worried the actual rate is below that. Please check before I need one.”

Use the beacons dataset.

Variable	Description
transmitted	Yes or No: whether the beacon transmitted successfully

Estimate the population success rate with a confidence interval and determine whether there is evidence that it is below 95%. Check the requirements for the method you use.

Your analysis and interpretation here.

Question 3: Did Quiet Hours Help?

Frank: “Remember quiet hours? We have sleep records from 30 crew members. Each person logged one rest cycle before the policy and one after it. The rows match: the two readings in a row belong to the same person. Did typical sleep increase?”

Use the quiet_hours dataset.

Variable	Description
crew_id	Crew member identifier
sleep_before	Hours slept before the quiet-hours policy
sleep_after	Hours slept after the quiet-hours policy

Determine whether typical sleep increased after the policy. Estimate the change with an appropriate confidence interval. Explain how the structure of these records affects your analysis and how you interpret “typical.”

Your analysis and interpretation here.

Question 4: Which Filter Should We Order?

Frank: “We tested two filter designs, Standard and Revised, on separate cartridges. Each cartridge got used once. I want to know whether their typical operating lifetimes differ. Command has a separate concern: is the Revised design more consistent? A longer lifetime and a more predictable lifetime are different things, apparently. I do understand that. Mostly.”

Use the `filters` dataset. There are 25 cartridges of each design.

Variable	Description
<code>design</code>	Standard or Revised
<code>lifetime_hours</code>	Hours until the cartridge needed replacement

Address both questions: do typical lifetimes differ, and is lifetime variability smaller for the Revised design? Estimate the difference in typical lifetime with an appropriate confidence interval. Explain what each analysis contributes to the purchasing decision.

Your analysis and interpretation here.

Question 5: Where Are Repairs Taking Longest?

Frank: “We pulled 60 separate repair tickets: 20 each for Reactor, Life Support, and Sensor Array. Command wants to know if typical repair time differs across those systems. If it does, tell us which systems differ. This is going into the official staffing recommendation, so be careful about declaring differences just because we asked several questions.”

Use the `repairs` dataset. Each ticket concerns a different repair event.

Variable	Description
<code>system</code>	Reactor, Life Support, or Sensor Array
<code>repair_hours</code>	Hours needed to complete the repair

Determine whether typical repair time differs across the three systems. Investigate specific differences if warranted. Justify your choices, including how you handle multiple comparisons.

Your analysis and interpretation here.

Question 6: Does the Surface Matter?

Frank: “Science tried Spray, Foam, and Gel on Hull Plating and Viewport Glass. There were ten separate samples for each treatment-and-surface combination, with samples randomly assigned to the treatment conditions. Does the relationship between treatment and containment depend on the surface? If it doesn’t, Command wants to know whether treatment or surface is associated with containment on its own.”

Use the surfaces dataset.

Variable	Description
treatment	Spray, Foam, or Gel
surface	Hull Plating or Viewport Glass
killed_pct	Percentage of Cinderspore killed on the sample

Answer Frank’s question and pursue the follow-up he requests if appropriate. Include a graph that helps explain the relationship. You do not need to identify specific pairwise differences for this question.

Your analysis and interpretation here.

Question 7: Are Some Roles Struggling More?

Frank: “Fleet Medical sent current wellness reports from 160 different crew members. Each person has one job role and one wellness classification. Is wellness related to job role, or does the distribution of wellness look the same across the roles?”

Use the wellness dataset.

Variable	Description
crew_role	Command, Engineering, Science, or Medical
wellness_status	Thriving, Steady, Strained, or Critical

Summarize wellness within each role and determine whether wellness status is associated with job role. Check the requirements for your analysis. You do not need additional tests to identify individual categories that differ.

Your analysis and interpretation here.

Question 8: What Explains Oxygen Output?

Frank: “Last file. Forty separate scrubber runs. Airflow, filter age, humidity, oxygen output. I want to know how airflow and output move together. Command wants to go further: can these three measurements help explain output, and which ones are related to output when we account for the other two?”

Use the oxygen dataset.

Variable	Description
o2_output_lpm	Oxygen output (liters/minute)
airflow_lpm	Incoming airflow (liters/minute)
filter_age_hours	Hours of use on the filter

Variable	Description
humidity_pct	Relative humidity of incoming air (%)

Describe the direction and strength of the association between airflow and oxygen output using a graph and an appropriate numerical measure. Then construct a model for oxygen output using all three explanatory variables.

Assess the model's assumptions, write its estimated equation, and evaluate whether the model as a whole is useful for explaining output. Identify which explanatory variables are statistically significant after accounting for the others. Interpret the airflow coefficient and the proportion of variation explained by the model.

Your analysis and interpretation here.

Final Message to Mission Control

Write 150–200 words summarizing your most important findings. Recommend two actions supported by your results, and identify one limitation Mission Control should consider. You do not need to repeat every test statistic or p -value.

Your message here.